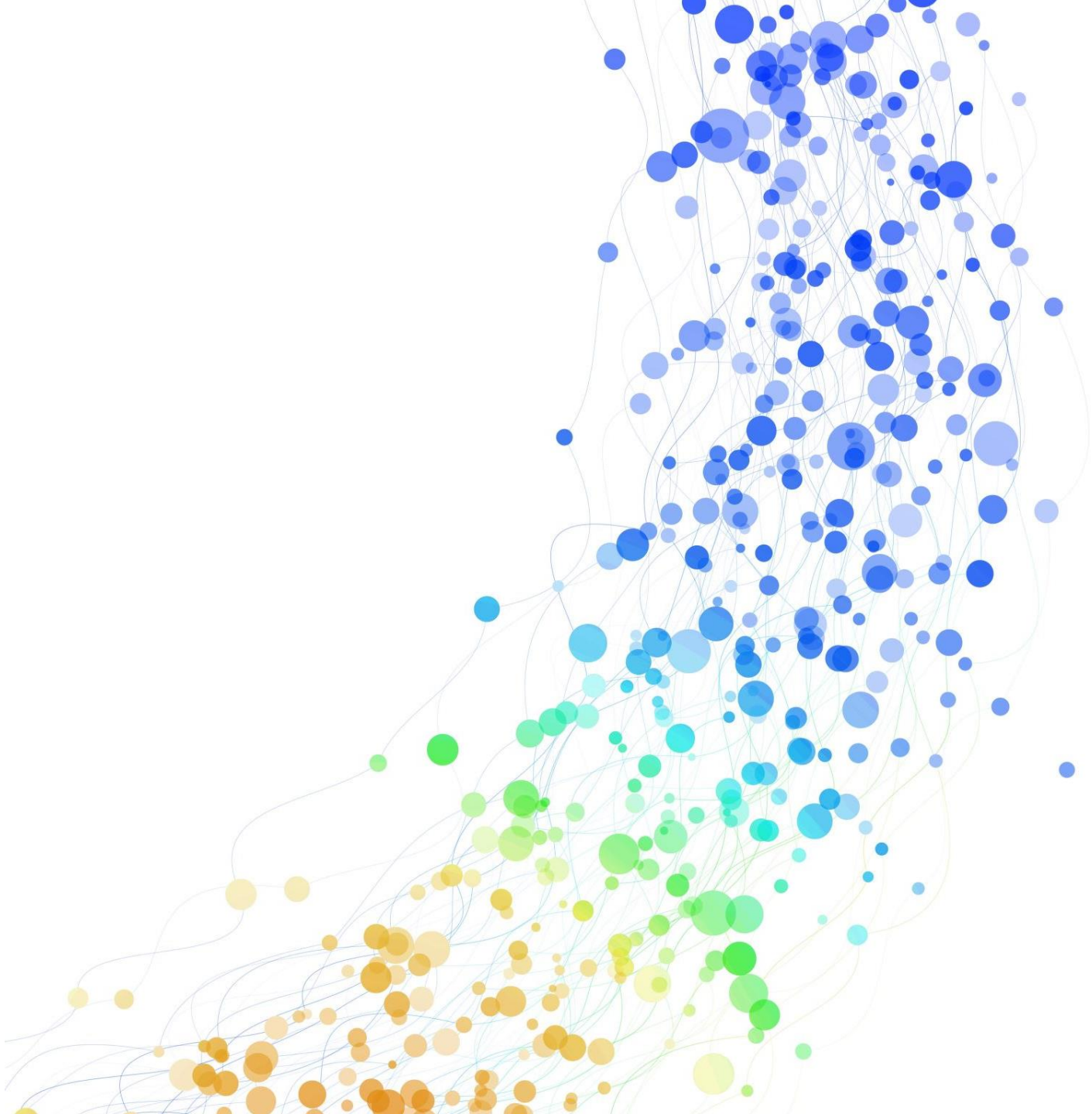
Predicating Shipping Prices in E-Commerce

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Goals



HOW DOES SHIPPING WORK WITHIN THIS COMPANY



CAN WE PREDICT SHIPPING COST USING ML MODEL



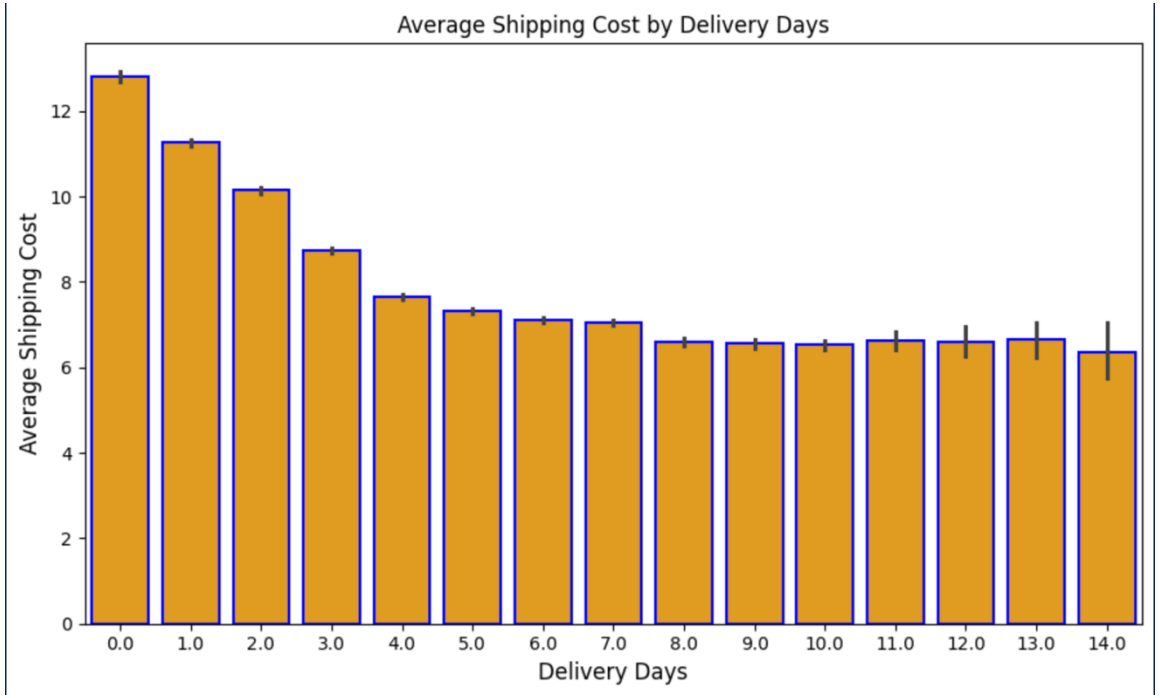
WHAT FEATURES ARE IMPORTANT IN ESTIMATING SHIPPING COST

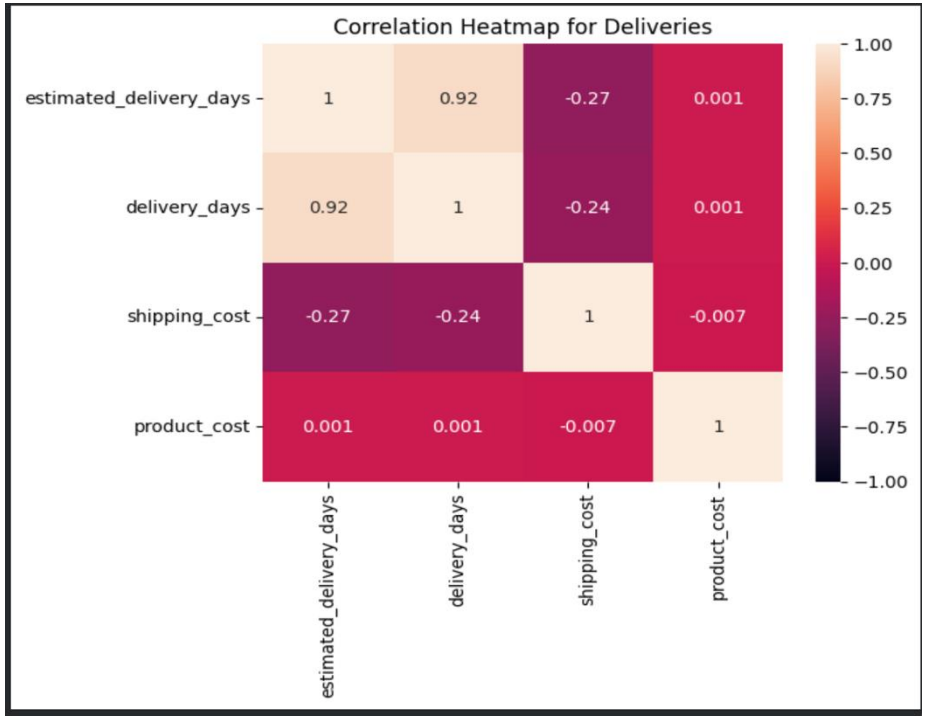
Hypothesis

Delivery days/estimated delivery days are important for predicating shipping cost

As delivery days increases the shipping cost will decrease


Distribution of Shipping Costs
Frequency
Shipping Cost
0
20000
40000
60000
80000
100000
120000
140000
0
5
10
15
20
25
30
35


Average Shipping Cost by Delivery Days
Average Shipping Cost
Delivery Days
0
2
4
6
8
10
12
0.0
1.0
2.0
3.0
4.0
5.0
6.0
7.0
8.0
9.0
10.0
11.0
12.0
13.0
14.0


Correlation Heatmap for Deliveries
estimated_delivery_days
delivery_days
shipping_cost
product_cost
1
0.92
-0.27
0.001
0.92
1
-0.24
0.001
-0.27
-0.24
1
-0.007
0.001
0.001
-0.007
1
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

The Model

Linear Regression on shipping cost

Features used: estimated delivery days, delivery days, customer segment, tax amount, shipping method, customer rating, discount amount, gross sales, customer lifetime value, product cost, profit, warehouse, delivery status, customer order count, net sales

How Did The Model Perform

- R2 = 1.0
- Suspected overfitting
- Investigated if there was an exact mathematical relationship involved
- MAE = 0

Correlation

	shipping_cost
estimated_delivery_days	-0.268131
delivery_days	-0.243900
customer_segment	-0.167375
tax_amount	-0.044784
shipping_method	-0.025003
customer_rating	-0.022041
discount_amount	-0.018133
quantity	-0.014454
gross_sales	-0.007107
customer_lifetime_value	-0.007026
product_cost	-0.006931
profit	-0.006290
warehouse	-0.002804
delivery_status	-0.001109
customer_order_count	0.000389
net_sales	0.003967

LR Coefficients

Features	Coefficent
net_sales	1.000000e+00
discount_amount	4.000000e-01
quantity	6.579311e-14
customer_segment	5.127062e-14
estimated_delivery_days	9.723078e-16
customer_lifetime_value	2.818926e-17
warehouse	-4.765479e-16
shipping_method	-8.218017e-16
delivery_status	-1.273016e-15
delivery_days	-3.087645e-15
customer_rating	-1.873415e-14
customer_order_count	-4.231678e-14
tax_amount	-4.000000e-01
gross_sales	-4.000000e-01
product_cost	-6.000000e-01
profit	-6.000000e-01

Model without net sales: R2 = 0.11

Measure of Feature Importance

Features	Importance
net_sales	16156.077472
gross_sales	3296.915575
product_cost	2555.737026
profit	1009.839064
discount_amount	280.588209
tax_amount	34.640510
customer_rating	0.000000
customer_segment	0.000000
delivery_status	0.000000
shipping_method	0.000000
delivery_days	0.000000
estimated_delivery_days	0.000000
warehouse	0.000000
customer_lifetime_value	0.000000
customer_order_count	0.000000
quantity	0.000000

- There seems to be an exact mathematical formula that calculates shipping cost



Conclusion



Succeeded in making a model that can calculate shipping cost



If model was given new test data it would predict shipping cost perfectly



Failed in achieving the desired outcome: model mathematically calculates



Model not always needed when given enough information